Subject:	OOP-JAVA
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Practical-14

GIT REPOSITRY LINK :

https://github.com/25aiml013-sudo/JAVA-PRACTICALS.git

Part A

PROBLEM :

1. Assemble and test: build a tiny layered app (a model, a service using an ArrayList, and a menu main), then write two or three JUnit @Test methods checking that adding increases the count and an invalid action throws — practice before testing MiniBank.

2. (For Advanced Learners – home) Organise that app into packages, build a runnable JAR, and add a test that saves and reloads its data unchanged.

Command :

```
PS D:\colleg\JAVA\JAVA-Practical> cd .\MiniBank\
PS D:\colleg\JAVA\JAVA-Practical\MiniBank> javac -d out src\annotation\*.java src\exception\*.java src\model\*.java src\repository\*.java src\util\*.java src\concurrency\*.java src\service\*.java src\*.java
PS D:\colleg\JAVA\JAVA-Practical\MiniBank> jar cfe minibank.jar MiniBank -C out .
PS D:\colleg\JAVA\JAVA-Practical\MiniBank>
```

Output :



Key Questions :

1. What is layered architecture (Model, Service, UI)?

- Model: Stores application data.
- Service: Contains business logic.
- UI: Interacts with the user.
- It makes code easier to maintain and test.

2. How do you test exceptions in JUnit?

- Use assertThrows() to check whether a method throws the expected exception.
- The test passes if the expected exception occurs.

3. What is a runnable JAR?

- A JAR packages compiled Java code into a single file.
- It can be executed using java -jar filename.jar.